

TAYLOR'S VERSION

FALL 2023

Econ 0100 | MiniExam Z

This MiniExam will take 15 minutes with quick break to follow. MiniExams are designed to both test your knowledge and challenge you to apply familiar concepts in new environments. Treat it as if you're trying to show me that you understand the material.

Academic Conduct Code

The following academic conduct code is designed to protect the integrity of your work. Print your initials beside the four academic honesty agreements before beginning.

I pledge to my fellow students, the university and the instructor...

... I _____ will complete this MiniExam solely using my own work

... I _____ will not use any internet connected devices or other online resources

... I _____ may use hardcopy resources (the textbook, printed materials, hardcopy notes)

... I _____ will not communicate with others during the MiniExam

Glittering Gum

Hogsmeade has recently seen a surge in shops selling Glittering Gum, a magical candy that makes the consumer's words sparkle and glisten as they speak. The Demand (marginal benefit) curve and Supply (marginal cost) curve for Glittering Gum can be represented using the following relationships:

$$D : P_b = 60 - Q_d$$

$$S : P_s = 10 + Q_s$$

However, a magical externality has arisen: the widespread chewing of Glittering Gum has led to an increase in "Starlight Sneeze," a condition where non-consumers start sneezing out harmless but bothersome glitter. This condition imposes an inconvenience cost of 10 Galleons to both consumers and non-consumers in Hogsmeade.

Question 1 (of 3) | Market Equilibrium

Use a graph and algebra to find equilibrium price, quantity, and DWL. Be sure to show your work. (continued below)

$$MB = MC$$

$$60 - Q = 10 + Q$$

$$50 = 2Q$$

$\hat{Q} = 25$
$\hat{P} = 35$

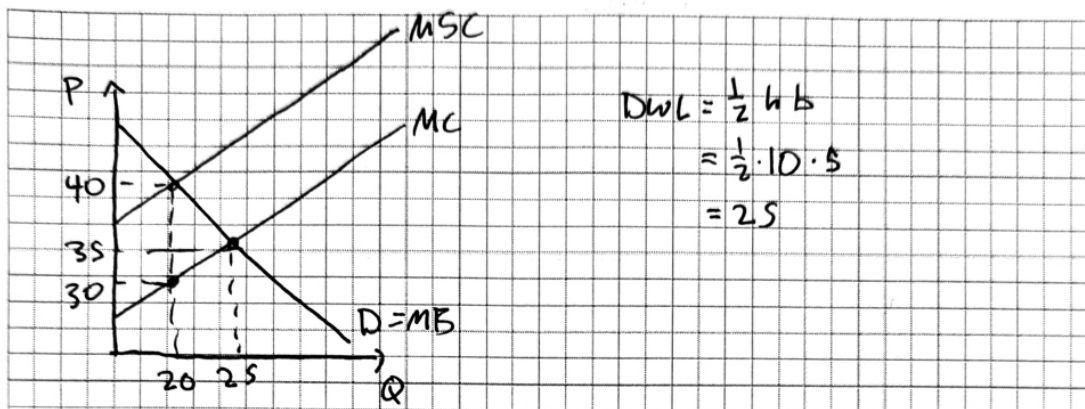
$$MSB = MSC$$

$$60 - Q = 10 + Q + 10$$

$$40 = 2Q$$

$\hat{Q} = 20$

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 ...continued.



Market Equilibrium Price: 35 Market Equilibrium Quantity: 25
 Socially Efficient Quantity: 20 Deadweight Loss: 25

Question 2 (of 3) | Policy Proposal

What type and size of policy would you propose to eliminate the DWL you identified in Question 1?

Policy Type: TAX Policy Size: 10
 Buyer Price: 40 Seller Price: 30 Post-Policy Quantity: 20

Question 3 (of 3) | Deadweight Loss Intuition

If the marginal cost of a gallon of gas is \$4.00, the negative externality is \$2.50, and the marginal benefit is \$4.50, what is the deadweight loss?

Deadweight Loss: 2

$$DWL = MSB - MSC$$

$$= 4.50 - 4.00 - 2.20$$

$$= -2$$